



Fusion of bidirectional neural networks decision for a high accuracy breast cancer detection based on DNA biomarker

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ABSTRACT

This study investigates the potential of deep learning-based sequence modeling for breast cancer classification using DNA biomarker data. A decision fusion method Bidirectional Long Short-Term Memory and Gated Recurrent Unit is proposed to distinguish between normal and cancer-related protein sequences. A total of 1217 sequences were collected from the Protein Data Bank, pre-processed using multiple alignment of fast Fourier transform and basic local alignment search tool for similarity analysis, and transformed into numerical representations using integer and label encoding. The results demonstrate that the rule-based decision fusion approach has effectively classified DNA biomarkers with an accuracy rate of 99.18%.

1. Introduction

Breast cancer is the most prevalent type of cancer among women worldwide, accounting for approximately 25% of all cancer diagnoses and remaining the leading cause of cancer-related mortality, contributing to 16% of such deaths in 2023 [1]. One approach to detect cancer involves the analysis of DNA biomarkers, as cancer typically originates from abnormal cellular proliferation resulting from genetic alterations [2]. Mutations in the BRCA1 and BRCA2 genes have been shown to significantly increase the risk of breast cancer and contribute to chemotherapy resistance, thereby compromising treatment efficacy [3]. These genes function as tumor suppressors and play a critical role in repairing DNA damage; their mutations can lead to deregulated cell division, ultimately elevating cancer risk [4].

Advancements in data acquisition and sequencing methodologies have driven significant progress in the bioinformatic analysis of DNA biomarker data [5]. However, the complex and high-dimensional nature of genetic data necessitates more sophisticated analytical approaches [6]. In this context, deep learning (DL) has emerged as a powerful tool, particularly in handling sequential data such as nucleotide and amino acid sequences [7].

Several studies have demonstrated the potential of DL in cancer detection using DNA and protein biomarkers. For instance Chen et al. [8] developed an innovative method for classifying breast cancer biomarkers (Ki67 and NHG) based on RNA sequencing data by transforming the sequences into artificial image objects, which were then classified using

convolutional neural networks (CNNs). Their model achieved an accuracy of 82.1% and an AUC of 0.891 for Ki67, and 82.0% accuracy with an AUC of 0.931 for NHG, outperforming manual pathologist assessments. Roslidar et al. [9] applied a Bidirectional Long Short-Term Memory (BiLSTM) model to classify protein sequences from the Protein Data Bank (PDB), encompassing 250 classes, and achieved 99% accuracy with a loss value of 0.042. Meanwhile, Jha et al. [10] employed a Bidirectional Gated Recurrent Unit (BiGRU) architecture combined with multimodal learning to predict human protein–protein interactions, attaining an accuracy of 98.29%, surpassing prior methods.

Recent studies published in 2022 have strengthened the role of DL in biomarker-based cancer analysis. Rukhsar et al. [11] analyzed RNA-Seq gene expression data for five cancer types and evaluated eight DL algorithms, concluding that CNN provided the most accurate and stable performance for multi-tumor classification. Similarly, Divate et al. [12] proposed a DL-based pan-cancer classification model using transcriptomic data from 37 cancer types, achieving 97% accuracy and identifying 976 tissue specific gene signatures through SHAP-based interpretability analysis. In the specific context of breast cancer, Shah et al. [13] introduced an ensemble framework combining LSTM, GRU, and BiLSTM architectures for detecting adenocarcinoma mutations from DNA sequences, achieving 99.57% accuracy and an AUC of 1.0.

Although various biomedical data, such as RNA-seq, gene expression, and visual representations, have been widely applied in breast cancer detection, the direct application of deep learning architectures,

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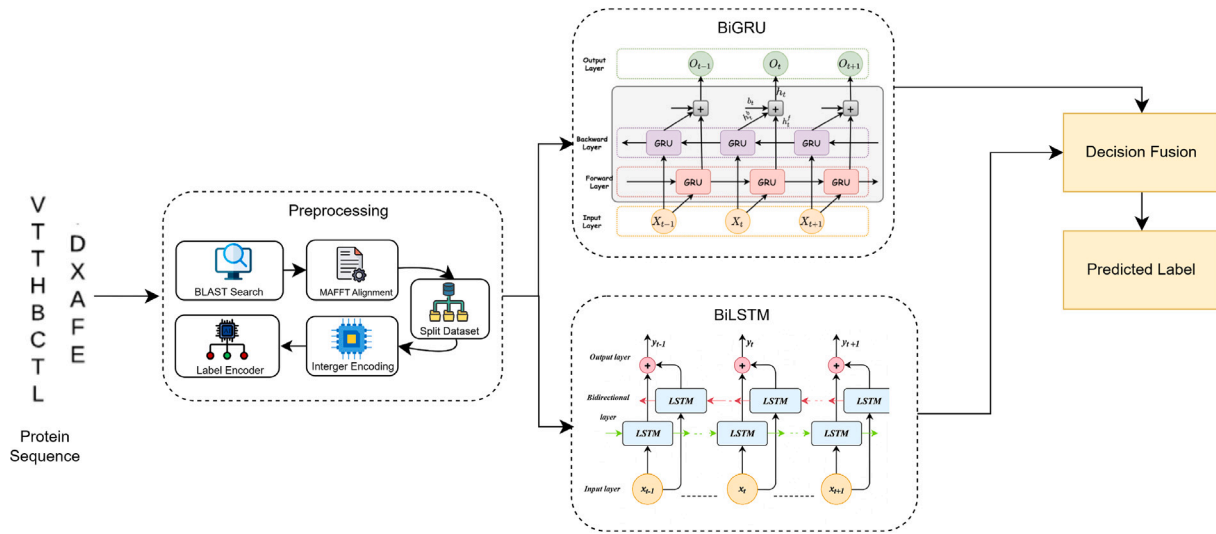


Fig. 1. Proposed framework.

including recurrent neural networks (RNNs), to DNA biomarker data based on protein sequences remains relatively limited. This study aims to address this gap by developing a DL model using two of the bidirectional neural network architectures, namely BiLSTM and BiGRU, in classifying protein sequences. The state-of-the-art approach in our work is the decision fusion technique, which aims to improve the accuracy rate and has not been previously proposed for recognizing protein biomarkers, particularly in breast cancer detection. In short, the main contributions of this article are as follows:

1. Developing a DNA biomarker classification model based on Recurrent Neural Network (RNN) architectures, based on BiGRU and BiLSTM.
2. Utilizing a protein sequence dataset as the basis for carcinogenesis by identifying structural regions and physiologically significant sequence patterns that differentiate normal from cancer-related proteins.
3. Proposing a novel rule-based decision fusion approach to integrate BiLSTM and BiGRU predictions, using the prior knowledge of the learning results.
4. Providing a comparison of the proposed fusion decision RNN model with similar existing models on the DNA biomarker dataset.
5. Presenting a discussion on the biological interpretation of protein sequence data used in this study to allow future challenges in developing effective protein sequence data alignment algorithms, considering the complexity.

The rest of this article is organized as follows: Section 2 presents the proposed methodology. Section 3 discusses the simulation results. Finally, Section 4 concludes the study.

2. Material and methods

The proposed framework is illustrated in Fig. 1. The process begins with the acquisition of protein biomarker datasets from the Protein Data Bank (PDB), followed by data preprocessing using the Basic Local Alignment Search Tool (BLAST), Multiple Alignment using Fast Fourier Transform (MAFFT) [14] alignment, and numerical encoding. Two RNN models, namely BiLSTM and BiGRU, were constructed with hyperparameter tuning to achieve optimal performance. Finally, a rule-based decision fusion method was implemented to enhance classification accuracy.

Table 1

DNA biomarkers of normal breast.

Name	Gene	Number of proteins
Breast Cancer Metastasis Suppressor	BRMS	18
Prolactin	PRL	31
Casein	CSN	80
Estrogen	ESR	100
Progesterone	PGR	100
Tumor Protein Suppressor	TP53	100
Retinoblastoma	RB1	100
RAD51	RAD51	81
Total protein of normal breast		610

2.1. DNA biomarker dataset

The dataset include amino acid sequences of genes that serve as biological markers (biomarkers) associated with either healthy or abnormal breast tissue obtained from the PDB (<https://www.ncbi.nlm.nih.gov/>). The detailed information on the genes employed as biomarkers for each class is presented in Tables 1 and 2. In total, the dataset comprises 610 protein sequences for normal breast tissue and 607 protein sequences for cancerous tissue, yielding 1217 protein sequences. Each sequence varies in length depending on its originating gene. Since deep learning models require uniform input dimensions, sequence alignment was performed to standardize lengths and synchronize residue positions before encoding.

Gene selection was informed by biological and clinical references that demonstrate direct associations between gene expression and breast cancer risk or protection. For instance, BRCA1 and BRCA2 genes are critical for DNA damage repair, while HER2 is a well-known therapeutic target due to its frequent amplification in breast cancer. Conversely, genes such as PRL, ESR, and TP53 serve as biomarkers of normal breast tissue physiology, indicating functions such as hormone production, cell cycle regulation, and protection against genetic mutations [15,16].

2.2. Data preprocessing

As shown in Fig. 1, the data preprocessing comprises of sequence similarity and alignment, dataset partition, and encoding. In detail, the data preprocessing steps are as follows:

1. Sequence Similarity and Alignment. BLAST (Basic Local Alignment Search Tool) was applied to identify sequence similarities with reference data. Alignment was then performed using

Table 2
DNA biomarkers of cancer breast.

Name	Gene	Number of proteins
Breast Cancer Gene 1	BRCA1	100
Breast Cancer Gene 2	BRCA2	100
Human Epidermal Growth Factor Receptor 2	HER2	100
Cortactin	CTTN	100
Marker of Proliferation	MK167	68
KI-67		
AKT1	AKT1	95
Mucin 1	MUC1	44
Total protein of cancer breast		607

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>3PXE_A Chain A, Breast cancer type 1 susceptibility protein [Homo sapiens]
VNRKSHMVSGLTPEEFMLVYKFAKHITLTNLITETTHVMKTDAEFVCERTLKLYFLGIAGGKVVSVFWTQSIKE
RKLMLNEHDFEVRGDIVNGRNHQGPKRARESDQRKIFRGLIECCYGPFTNMPDQLEWVQLCGASVVKELSSFTLGTGVH
PZVVQPDANTEDNGFHAIGQCEAPVTRKVVLDVVALYQCELDLYLIPQIP
>NP_001395442.1 breast cancer type 1 susceptibility protein isoform 146 [Homo sapiens]
MSLQESTRFSQLVEELKIIICAFQDITGLVANSYNFAKKENNSPEHLKDEVSIIQSMGYRNRAKRLQSEPNPSLQET
SLSVQLSNLGTVRTLRTRKIQPKTSVYIELGSDSSEDVTNATYCSVDGQELLQITPQGTDRDEISLDSAKKAACEFSE
TDVNTIEHHQPSNNDLNTTEKRAAERHPEKVGGAASGCESETSVESGSLSSQSDIITQRDTMQHNLIKLQQAEL
EAVLQHGSPNSYPSITISDSALIDLNPQSTSEKGTPLYESGSLFSDDPESDSEDRAPEASARVGNIPSSSAL
KVPQLKVAESAQAHAHTTDTAGYNAMEESVREKPELTASTERNVNRHSHVSVGLTPEEFMLVYKFAKHITLTNLII
TEETHVMKTDAEFVCERTLKLYFLGIAGGKVVSVFWTQSIKERKMLNEHDFEVRGDIVNGRNHQGPKRARESDQRKI
FRGLIECCYGPFTNMPDQLEWVQLCGASVVKELSSFTLGTGVHPZVVQPDANTEDNGFHAIGQCEAPVTRKVVLD
SVALYQCELDLYLIPQIPHSY
>KAIZ583239.1 BRCA1 DNA repair associated [Homo sapiens]
HMSCGSLQNNVPSQELIKVVDVEEQLEESGPHDLTETSVIPQDLEGTPYLESGISLFSDDPESDSEDRAPEASAR
VGNIPSSSALVKPQLKVAESAQAHAHTTDTAGYNAMEESVREKPELTASTERNVNRHSHVSVGLTPEEFMLVYKFA
RKHITLTNLITETTHVMKTDAEFVCERTLKLYFLGIAGGKVVSVFWTQSIKERKMLNEHDFEVRGDIVNGRNHQG
KRAESDQRKIFRGLIECCYGPFTNMPDQLEWVQLCGASVVKELSSFTLGTGVHPZVVQPDANTEDNGFHAIGQCE
APVTRKVVLDVVALYQCELDLYLIPQIPHSY

```

Fig. 2. Example of DNA biomarker data prior to BLAST processing.

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-----SLPGFRN-----LHIDDQIT-----LIQYSWMS--
-----SLPGFRN-----LHIDDQIT-----LIQYSWMS--
-----SLPGFRN-----LHIDDQIT-----LIQYSWMS--
-----SLPGFRN-----LHIDDQIT-----LIQYSWMS--
-----SLPGFRN-----LHIDDQIT-----LIQYSWMS--
-----SLPGFRN-----LHIDDQIT-----LIQYSWMS--

```

Fig. 3. Result of DNA sequence alignment using MAFFT.

MAFFT (Multiple Alignment using Fast Fourier Transform) with the G-INS-i algorithm, selected for its high accuracy despite requiring longer computation due to the use of a default guide tree [17]. Alignment ensured homologous residues were positioned in the same columns across sequences. Visualizations before and after alignment are shown in Figs. 2 and 3.

- Dataset Partition. The dataset was divided into training (80%), validation (10%), and testing (10%) subsets to enable objective performance evaluation.
- Encoding. The aligned sequences were converted into numerical form using integer encoding and label encoding, making them suitable for neural network input.

The aligned sequences were converted into numerical form using integer encoding and label encoding, making them suitable for neural network input. To ensure uniform input dimensions for the deep learning models, sequences shorter than the maximum length of 10,840 amino acids were padded with zeros at the end (padding='post'), while longer sequences were truncated to maintain consistency. After encoding and embedding into a 128-dimensional vector space, each sequence was represented as a fixed-size input tensor of shape 10,840 × 128.

2.3. Bidirectional neural network model development

BiLSTM is an extension of LSTM [18] that processes sequential data in both forward and backward directions. Each input sequence is fed into two separate LSTM networks: one processes the sequence from beginning to end, while the other processes it from end to beginning.

Their outputs are concatenated and passed to the output layer. This approach provides the network with the full context of the entire sequence, both preceding and succeeding a specific time step, thereby improving accuracy in tasks that require global contextual understanding [19]. Meanwhile, BiGRU is an extension of GRU [20,21] that processes data in two directions, forward and backward. Accordingly, the final hidden representation in BiGRU is a combination of the forward hidden state and the backward hidden state [22].

Both BiLSTM and BiGRU architectures were implemented using pretrained embeddings. The detailed configuration of these models is summarized in Table 3. As shown in the table, each model consists of an input layer that receives the sequence data, an embedding layer initialized with pretrained weights to represent input features, a bidirectional recurrent layer (either BiLSTM or BiGRU) with 256 output units, a dropout layer to prevent overfitting, and a dense layer with a sigmoid activation function for binary classification. The number of parameters differs between BiLSTM and BiGRU because of the distinct internal gating mechanisms in each architecture. The complete pseudocode of the proposed protein sequence fusion decision process combining BiGRU and BiLSTM is presented in Algorithm 1.

2.4. Hyperparameter

Model training was implemented using the Tensorflow framework, with hyperparameters optimized through iterative experimentation. Several parameter configurations were tested, and the final settings are summarized in Table 4. Key parameters include optimizer type, learning rate, batch size, and number of epochs. The selected configuration was chosen to maximize performance while avoiding overfitting.

Algorithm 1 Pseudocode of Proposed DNA Biomarker Fusion Decision of BiGRU and BiLSTM Process

Require: Protein sequences in FASTA format (*Normal, Cancer*)
Ensure: Final trained BiGRU and BiLSTM models with evaluation metrics

- 1: Load protein sequences from FASTA files
- 2: Remove gaps ('-') and compute amino acid frequencies
- 3: Encode amino acids into integers using a predefined mapping
- 4: Pad sequences to a uniform *max_length*
- 5: Split dataset: 80% train, 10% validation, 10% test
- 6: Encode class labels (*normal, cancer*) numerically
- 7: **for** each *model_type* in {BiGRU, BiLSTM} **do**
- 8: Define model:
 Input → Embedding(128) → Bidirectional(RNN(128)) → Dropout(0.2)
 → Dense(1, sigmoid)
 where RNN ∈ {GRU, LSTM}
- 9: Compile with Adam (learning rate ∈ {0.001, 0.005, 0.0001, 0.0005}) and binary cross-entropy
- 10: Train models with epochs ∈ {50, 75, 100} (batch_size = 32) using ModelCheckpoint(val_loss)
- 11: Select best model based on highest validation F1-score:
 BiLSTM best at lr = 0.0001, epochs = 75
 BiGRU best at lr = 0.005, epochs = 100
- 12: Evaluate selected model on test set: compute accuracy, precision, recall, and F1-score
- 13: **end for**
- 14: **Decision Fusion (Averaging / Soft Voting):**
- 15: Predict probabilities for test set using BiLSTM (*prob_bilstm*) and BiGRU (*prob_bigru*)
- 16: Compute average probability: *avg_prob* = (*prob_bilstm* + *prob_bigru*)/2
- 17: Convert to binary predictions: *pred_fusion* = (*avg_prob* > 0.5)
- 18: Evaluate fusion predictions: compute accuracy, precision, recall, F1-score, specificity, AUC
- 19: Visualize:
 Amino acid frequency distribution
 Training/validation accuracy and loss curves
 Confusion matrices
- 20: **Return:** Final trained models (BiGRU and BiLSTM), fusion predictions and their evaluation metrics

Table 3
Architecture of BiLSTM and BiGRU models.

Layer (type)	Output shape	BiLSTM param #	BiGRU param
InputLayer (input_layer)	(None, 10840)	0	0
Embedding (embedding)	(None, 10840, 128)	2688	2688
Bidirectional (bidirectional)	(None, 256)	263,168	198,144
Dropout (dropout)	(None, 256)	0	0
Dense (dense)	(None, 1)	257	257

Table 4
Hyperparameter.

Hyperparameter	Value
Optimizer	Adam
Batch Size	32
Learning Rate	0.001, 0.005, 0.0001, 0.0005
Embedding	128
BiLSTM/BiGRU layer	128
Activation Function	Tanh, Sigmoid
Max Epoch	50, 75, 100
Dropout Rate	0.2
Loss Function	Binary Cross Entropy
Checkpoint	Validation loss

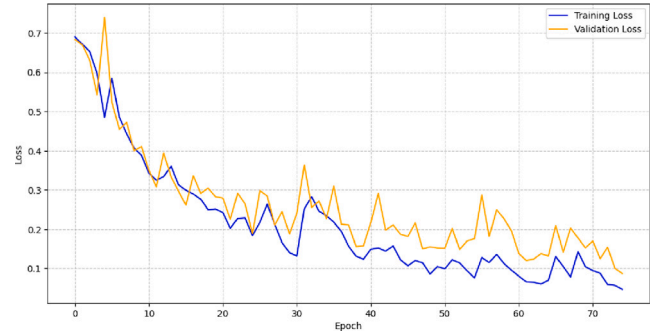


Fig. 4. Learning curve of the BiLSTM model with learning rate 0.0001 over 75 epochs.

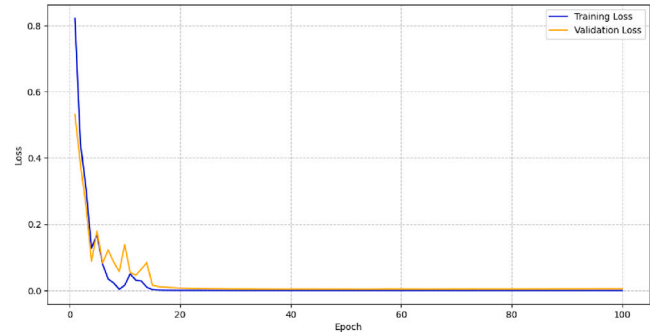


Fig. 5. Learning curve of the BiGRU model with learning rate 0.005 over 100 epochs.

2.5. Decision fusion rules

After observing the learning and testing results, we found that both models (BiLSTM and BiGRU) exhibit strong generalization ability for normal protein sequence data. Thus, taking the average of the probability prediction from both models is an effective decision fusion mechanism. This approach smooths out model-specific fluctuations, providing a more stable final prediction compared to using each model individually. The fusion formulation is defined as follows:

$$\hat{y}_{\text{fusion}}(i) = \begin{cases} 1, & \text{if } \frac{p_{\text{BiLSTM}}(i) + p_{\text{BiGRU}}(i)}{2} > 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where $p_{\text{BiGRU}}(i)$ and $p_{\text{BiLSTM}}(i)$ are the predicted probabilities generated by the BiGRU and BiLSTM models for sample i , respectively, and $\hat{y}_{\text{fusion}}(i)$ is the final predicted label after averaging the probabilities from both models and applying a threshold of 0.5, for 1 as Cancer and 0 as Normal classes.

3. Result

3.1. Learning performance

During the training process, monitoring the loss value served as an important indicator for assessing the model's ability to learn the sequential structure of the data. In this study, a series of experiments were conducted using various learning rates and epochs. Among all combinations, the best learning curve of BiLSTM and BiGRU was obtained with a learning rate of 0.0001 and 75 epochs and 0.005 and 100 epochs, respectively as shown in Figs. 4 and 5. The curves show a gradual decrease in training and validation loss, with minor early fluctuations, but overall illustrates a stable learning process without clear signs of overfitting.

3.2. Evaluation metrics

The best model of BiLSTM and BiGRU were evaluated on the testing dataset to assess their generalization capability. Both Models achieved 98.36% accuracy, 97.14% precision, 100% recall, 98.55% F1-score, 96.30% Specificity, and 98.15% AUC. Despite their strong performance, each model still produced misclassifications on certain samples, as shown in Fig. 6, where two sequence data were predicted incorrectly. The BiLSTM model produced two false positives at indices 13 and 96,

while the BiGRU model produced two false positives at indices 66 and 96, indicating that sample 96 was consistently misclassified by both models.

The evaluation results showed that this decision fusion achieved 99.18% accuracy, 98.55% precision, 100% recall, 99.27% F1-score, 98.15% specificity, and 99.95% AUC. These values indicate an improvement of approximately 0.82% in accuracy compared to single models. Meanwhile, the confusion matrix in Fig. 7 demonstrates excellent performance, with only one remaining false positive after fusion, corresponding to sample index 96, which was misclassified by both base models. This finding confirms that the averaging-based fusion method can effectively reduce specific types of misclassifications, particularly false positives, thereby producing more consistent and reliable predictions.

3.3. Discussion

In addition to evaluating classification performance, this study also assessed model complexity and efficiency after the training process was completed. Measurements were conducted on the final model size and the training duration across various hyperparameter configurations.

Before training, the BiLSTM architecture consisted of 266,113 parameters (1.02 MB), whereas the BiGRU architecture contained 201,089 parameters (785.50 KB). Following training, the model sizes increased

Table 6
Performance comparison of existing recurrent neural network model for data sequence classifier.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Specificity (%)	AUC (%)
Transformer	95.90	94.37	98.53	96.40	92.59	95.56
ProtBERT	55.74	55.74	100	71.58	0	50.00
DNABERT	55.74	55.74	100	71.58	0	50.00
BiLSTM	98.36	97.14	100	98.55	96.30	98.15
BiGRU	98.36	97.14	100	98.55	96.30	98.15
Proposed Model	99.18	98.55	100	99.27	98.15	99.95

In addition, the protein sequence alignment performed using the progressive G-INS-1 method in the MAFFT software also contributed to the observed errors. Inaccurate alignment results may lead to shifts in the positions of critical amino acid information. When this information is not semantically aligned, the model may learn from inputs that are not representative of the true biological patterns. An illustration of this misalignment is shown in Fig. 8, which highlights discrepancies in sequence positions that should have been matched. Such inaccuracies hinder the construction of optimal feature representations during training, thereby increasing the likelihood of misclassifications when the model is tested with new data.

Thus, further work is needed to improve the RNN model’s performance in recognizing and classifying DNA biomarkers. Some to mention are developing robust alignment algorithms for each distinctive biological interpretation; training RNNs with more datasets from various protein sequence dataset resources; and implementing regularization and decision fusion mechanisms based on learning.

4. Conclusion

The findings of this study demonstrate that sequence modeling approaches based on deep learning, particularly BiLSTM and BiGRU architectures, are capable of classifying protein biomarker sequences for breast cancer detection with high performance using a single, modest-sized PDB-derived dataset. Based on evaluations across various hyperparameter configurations, both models achieved 100% accuracy and F1 score in certain settings.

However, there was an error due to the presence of unknown characters (“X”) in protein sequences, as well as inaccuracies during the alignment stage, which contributed to performance degradation. To address the limitations of single models, this study employed a decision fusion method based on averaging predicted probabilities that combined predictions from BiLSTM and BiGRU, resulting in an improved accuracy rate of 99.18%, with a precision of 98.55%, recall of 100%, and F1 score of 99.27%, specificity of 98.15%, and AUC of 99.95%. In addition to classification performance, both architectures demonstrated computational efficiency, with average training durations of less than 65 min for 100 epochs. These results reinforce the potential of the proposed models for real-world breast cancer detection systems under time and resource constraints. This work was a preliminary but promising study. Future work may focus on developing more adaptive preprocessing strategies and on exploring learning-based fusion methods to improve accuracy further and enhance model robustness against broader data variability.

CRediT authorship contribution statement

Roslidar Roslidar: Writing – review & editing, Investigation, Funding acquisition, Conceptualization. **Qurrata A’yuni:** Writing – original draft, Software, Methodology. **M. Zulhamsyah:** Software, Methodology. **Fitri Arnia:** Writing – review & editing, Supervision. **Afnan Afnan:** Writing – review & editing. **Essy Harnelly:** Validation, Formal analysis, Data curation.

Ethical compliance and data licensing

The protein sequence data used in this study were obtained from the RCSB Protein Data Bank (PDB), which is an open-access and publicly available database. According to the PDB Data Usage Policy, all deposited data are freely available for research and educational purposes without restriction on usage or redistribution. All PDB data are released under the Creative Commons CC0 1.0 Universal (Public Domain Dedication) license, permitting unrestricted use, distribution, and reproduction. No human or animal subjects were directly involved in this study; therefore, no ethical approval was required. All data were accessed directly from the official PDB repository, ensuring transparency and traceability.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT to translate the manuscript. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that there is no conflict of interest in this paper.

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